MIL-R-55656(EL)
Amendment 2
18 March 1969

SUPERSEDING
AMENDMENT 1
26 February 1969

MILITARY SPECIFICATION

Radio Set AN/GRC-103( ) (V) (Basic)

This amendment forms a part of MIL-R-55656(EL), dated 15 February 1969.

1. Paragraph 2 FEDERAL SPECIFICATIONS:

   a. Delete:

      QQ-W-343 Wire, Electrical (uninsulated)

   b. Change "PPP-F-760" to "PPP-S-730"

2. Paragraph 2 MILITARY SPECIFICATIONS:

   a. Delete:

      MIL-C-17 Cables, Radio Frequency, Coaxial, Dual Coaxial, Twin Conductor, and Twin Lead

      MIL-C-3432 Cable and Wire, Electrical (Power and Control, Flexible, 300 and 600 volts)
      MIL-C-3702 Cable, Power, Electrical, Ignition, High Tension
      MIL-C-3849 Cord, Electrical (Tinsel)
      MIL-C-3883 Cord, Electrical (Audio Frequency)
      MIL-C-3884 Cord, Electrical (Short Lay)
      MIL-C-3885 Cable Assemblies and Cord Assemblies, Electrical (for use in Electronic, Communication and Associated Electrical Equipment)
      MIL-I-3930 Insulating and Jacketing Compounds, Electrical (for cables, cords, and wires)
      MIL-C-11060 Cables, Twisted Pair Internal Hookup, Unshielded

FSC 5820

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MIL-H-11996 Handset H-60( )/PT, Handset H-156( )/U and Handset H-165( )/U
MIL-C-13777 Cable, Special Purpose, Electrical, General Specification for
MIL-C-22931 Cable, Radio Frequency, Semirigid, Coaxial, Semi-air-dielectric, General Specification for

b. Add:

MIL-B-23071 Blower, Miniature, for Cooling Electronic Equipment (10 to 500 CFM) General Specification for

3. Paragraph 2 STANDARDS:

a. Delete:

MIL-STD-16 Electrical and Electronic Reference Designations

b. Delete:

USAMCOM MDD-STD-1 Microelectronics (Integrated Electronics) and Modular Design in Electronic Equipment

4. Paragraph 2 PUBLICATIONS:

a Delete: DEPARTMENT OF THE ARMY

TM-11-5820-540-12 Operator and Organizational Maintenance Manual-Radio Set AN/GRC-103(V)

b. Delete: DEFENSE SUPPLY AGENCY

MIL-HDBK-216 RF Transmission Lines and Fittings (Application for copies should be addressed to the Commanding Officer, US Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Penna. 19120)

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5. Paragraph 3.4 Parts, materials and processes. Line 4: Delete "except for the following"

Subparagraph a: Delete.
Subparagraph b: Plating and coating. Delete.
Subparagraph c: One-part PC connector. Delete.

6. Paragraph 3.21.5 Dripproof. Delete.

7. Paragraph 3.21.6 Salt fog. Delete.

8. Paragraph 3.21.7 Sand and dust. Delete.

9. Paragraph 3.21.8 Fungus. Delete.

10. Paragraph 3.23 Electromagnetic interference characteristics. Delete and substitute:

"3.23 Electromagnetic interference characteristics. The equipment shall meet the electromagnetic interference requirements of MIL-STD-461A, Table A-1, as defined herein. When tested for cable and case leakage and antenna conducted interference, the performance of the model shall govern. The following class CE requirements apply to the AN/GRC-103( ) (V):

CE06 Key-up only CS01 RE02 RS03 A-CE1
CS06 A-CS1"

11. Table X - Group A inspection.

(C) Electrical (main assemblies) (page 51)
Dummy Load, Electrical (DA-437( )/GRC-103(V)

a. VSWR, Req. par. change "3.13.48" to "3.13.4.8"
b. Impedance, req. par. change "3.13.48" to 3.13.4.8"
c. Dissipation, req. par. change "3.13.48" to "3.13.4.8"

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12. Paragraph 4.7.1.1 Low temperature. Delete "-50°F" in fourth line and substitute "-35°F".

13. Paragraph 4.7.1.2 High temperature. Delete and substitute:

"4.7.1.2 High temperature. The equipment shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. The operating temperature shall be 125°F and the storage temperature shall be +155°F. Prior to, during, and after testing, the equipment shall meet full specification performance for the measurements specified in 4.9.1".

14. Paragraph 4.7.4.2 Shock test; ballistic. Line 11 delete "Table 3.8", and substitute "Table VIII".

15. Paragraph 4.7.3 Altitude (elevation) test. Delete and substitute:

"4.7.3 Altitude (elevation) test. The equipment shall be tested in accordance with Method 500, Procedure I, of MIL-STD-810, except the chamber internal pressure in step 2 shall be 282.0 mm of Hg (11.1 in. of Hg of 25,000 ft. above sea level). During and after testing, the equipment shall meet full specification performance for the measurements specified in 4.9.2. The chamber may be returned to ambient atmospheric pressure for frequency changes".

16. Paragraph 4.7.5 Drip-proof. Delete.

17. Paragraph 4.7.6 Salt fog test. Delete.

18. Paragraph 4.7.7 Dust test. Delete.

19. Paragraph 4.7.8 Fungus test. Delete.

20. Appendix III. Delete and substitute Appendix IIIB.

CUSTODIAN:

ARMY-EL

PREPARING ACTIVITY:
ARMY-EL

PROJECT NO. 5920-A417

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APPENDIX III B

DATED: 18 March 1969

10. SCOPE

10.1 This appendix covers the engineering changes that are not reflected in the model and shall be incorporated in the Radio Set AN/GRC-103( )(V) equipments produced under this specification. Changes shall be accomplished without impairing performance. Modifications are the contractor's responsibility and shall be submitted to the CODR for approval prior to incorporation into the equipment.

20. ENGINEERING CHANGES

20.1 This section describes the design characteristics of the model which are subject to change. Any difference between the model characteristics and the requirements of Section 3 represents the required engineering change.

20.1.1 Standard parts. An unknown number of non-standard parts are in the model. Their use does not constitute a waiver from the requirements of this specification or MIL-P-11268. The contractor shall propose a substitute conforming to MIL-P-11268, where a part, material, or process does not meet MIL-P-11268. Prior approval to use the proposed substitute shall be given by the CODR before the part, material or process is used.

20.1.2 Mast height. The mast was designed to meet the 50 foot height but has been tested only to 35 feet.

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